PRODUCT REQUIREMENTS SPECIFICATION

Chicken Egg Incubator Controller | FEARS Format | Version 1.0 | DRAFT

Document Title	Product Requirements Specification – Chicken Egg Incubator Controller
Product	Fully Automatic Egg Incubator (up to 100 eggs)
Format	FEARS (Functional / Electrical & Environmental / Appearance & Aesthetic / Regulatory & Reliability / Safety & Standards)
Target Market	Balkan Region (Serbia primary) and EU
Document Version	1.0 – Draft for Review
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Prepared By	Requirements Engineering Team (Electronics, Firmware, Mechanical, Regulatory)
Status	DRAFT – Pending technical and regulatory review

1. Introduction & Scope

This document defines the complete product requirements for a Fully Automatic Chicken Egg Incubator Controller in FEARS format. Requirements cover the full product lifecycle from manufacturing, first use, normal operation, edge and misuse conditions, maintenance, and end-of-life decommissioning. All requirements are written to support CE marking for the EU market and Serbian market compliance.

Priority notation: Must = mandatory requirement; Should = strongly recommended; May = optional / nice to have.

F | FUNCTIONAL REQUIREMENTS

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
F-001	Temperature Control	The system SHALL maintain egg chamber temperature in range 37.0°C – 39.0°C with a steady-state accuracy of ±0.1°C.	Must	Normal, First Use
F-002	Temperature Sensing	The system SHALL use a minimum of 2 independent temperature sensors (NTC/PT100) with cross-validation. If sensors disagree by >0.5°C, alarm shall be raised.	Must	Normal, Fault
F-003	Humidity Control	The system SHALL control relative humidity (RH) in the range 40%–80% RH ±3% RH via automatic water dispensing from external tank.	Must	Normal
F-004	Humidity Sensing	The system SHALL measure humidity using a digital RH/T sensor (e.g., SHT3x class). Sensor shall be recalibrated or cross-checked at each power-on.	Must	Normal, Startup
F-005	Water Level Monitoring	The system SHALL monitor the water tank level. An alert SHALL be displayed and audible alarm triggered when tank drops below minimum level.	Must	Normal, Depletion
F-006	Fan Operation	The system SHALL operate an internal circulation fan continuously during incubation to ensure uniform temperature distribution (≤±0.3°C variation chamber-wide).	Must	Normal

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
F-007	Fan Fault Detection	The system SHALL detect fan failure (stall or over-current) within 30 seconds and raise an audible + visual alarm.	Must	Fault
F-008	Egg Turning – Auto	The system SHALL automatically turn eggs at configurable intervals (default: every 2h ±5 min) through the turning mechanism, stopping 3 days before projected hatch.	Must	Normal
F-009	Egg Turning – Manual Override	The operator SHALL be able to manually trigger an egg turn from the touchscreen at any time outside the lockdown period.	Should	Manual Operation
F-010	Lockdown Mode	The system SHALL enter a 'Lockdown' mode in the last 3 days of incubation, stopping egg turning and adjusting humidity to 65%–70% RH automatically.	Must	Hatch Phase
F-011	Incubation Timer	The system SHALL track incubation day count from a user-set start date. Countdown to expected hatch date shall be displayed.	Must	Normal
F-012	LED Lighting	The system SHALL provide internal LED illumination, controllable via touchscreen (ON/OFF + brightness). LED SHALL automatically extinguish when door is opened.	Must	Normal, Door Open
F-013	Touchscreen Display	The system SHALL present real-time readings (temperature, humidity, fan status, water level, incubation day, alarms) on an LCD touchscreen ≥3.5 inches diagonal.	Must	Normal
F-014	Alarm System	The system SHALL provide audible (≥70 dB @ 1m) and visual alarms for: high/low temperature, high/low humidity, fan failure, water low, power failure, sensor fault.	Must	Fault, Emergency
F-015	Alarm Mute / Acknowledge	The user SHALL be able to acknowledge/mute alarms via touchscreen. Alarm condition shall be logged even if muted. Mute SHALL auto-expire after 60 minutes.	Must	Fault
F-016	Power Failure Handling	Upon power loss, the system SHALL retain all settings, alarm history, and incubation timer data in non-volatile memory (EEPROM/Flash). Upon restore, it SHALL resume operation automatically.	Must	Power Failure
F-017	Configurable Parameters	The operator SHALL be able to set target temperature, target humidity, turning interval, incubation duration, and alarm thresholds via the touchscreen menu.	Must	Setup
F-018	Factory Default Reset	The system SHALL provide a factory default reset function, accessible only through a multi-step confirmation on the touchscreen to prevent accidental reset.	Must	Setup, Edge Case
F-019	Heater Safety Cutoff	The system SHALL automatically disable all heaters if measured temperature exceeds 41.0°C (hard safety limit), regardless of software state.	Must	Safety, Fault
F-020	Self-Test on Power-Up	On each power-up, the system SHALL perform a self-test of sensors, heater, fan, and display, reporting results to the user before entering normal operation.	Must	Startup
F-021	Data Logging	The system SHOULD log temperature, humidity, alarm events and operator actions with timestamp, accessible via touchscreen log view (minimum 30-day retention).	Should	Normal, Audit
F-022	Child/Tamper Lock	The system SHALL provide a touchscreen lock function to prevent unintended parameter changes during active incubation.	Should	Edge Case, Misuse Prevention
F-023	Egg Capacity Indication	The system SHALL support up to 100 chicken eggs. Firmware SHALL NOT constrain egg count but documentation shall state maximum safe load.	Must	Normal

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
F-024	Language Support	The HMI (touchscreen) SHALL support at minimum English, Serbian, Croatian, and Romanian language settings.	Should	Balkan Market
F-025	Firmware Update	The system SHOULD support secure firmware updates via USB or SD card with integrity verification (CRC/hash check) and rollback capability.	Should	Lifecycle, Maintenance

E | ELECTRICAL & ENVIRONMENTAL REQUIREMENTS

ID	Parameter	Requirement Statement	Priority	Context
E-001	Supply Voltage	The device SHALL operate from 230V AC ±10%, 50 Hz mains supply (Balkan/EU standard).	Must	Normal
E-002	Power Consumption	Total system power consumption at full load (all heaters, fan, lights, controls) SHALL not exceed 200W.	Must	Normal
E-003	Mains Fusing	A suitably rated mains fuse (T2A min) SHALL be installed on the line conductor of the mains input, accessible for user replacement.	Must	Safety
E-004	Protective Earth	All metallic enclosure parts SHALL be connected to protective earth (PE). Earth continuity resistance SHALL be ≤0.1Ω.	Must	Safety, Regulatory
E-005	Heater Design	Heating elements SHALL be PTC or nichrome type, rated for ≥50,000 operating hours. Watt density SHALL be chosen to prevent surface temperature exceeding 80°C.	Must	Reliability
E-006	EMC – Emissions	The device SHALL comply with EN 55032 Class B conducted and radiated emission limits for residential/commercial use.	Must	Regulatory
E-007	EMC – Immunity	The device SHALL comply with EN 55035 immunity requirements. Critical functions (temperature control, alarms) SHALL remain operational during ESD, EFT/Burst, and surge tests.	Must	Regulatory
E-008	ESD Protection	All user-accessible interfaces (touchscreen, USB/SD card slot) SHALL be protected with TVS diodes to withstand ±4kV contact / ±8kV air ESD (IEC 61000-4-2 Level 2/3).	Must	Robustness
E-009	Surge Protection	The mains input SHALL include a MOV-based surge protector capable of absorbing 1kV/2kV differential/common mode surges per IEC 61000-4-5.	Must	Robustness
E-010	Operating Temperature (Electronics)	All electronic components SHALL be rated for continuous operation at ambient temperatures of 0°C to +50°C.	Must	Environmental
E-011	Storage Temperature	The device SHALL withstand storage at -20°C to +70°C without permanent damage (non-operational).	Must	Lifecycle
E-012	Humidity (Electronics)	Electronics SHALL be protected against condensation. PCBs SHALL be conformally coated or adequately enclosed. Operating RH for electronics: 20% – 85% non-condensing.	Must	Environmental
E-013	Ingress Protection – Water System	Components in contact with water (humidity system, tank sensors) SHALL be rated IP54 minimum. Main electronics enclosure: IP22 minimum.	Must	Safety, Environmental

ID	Parameter	Requirement Statement	Priority	Context
E-014	Insulation Resistance	Insulation resistance between mains-live conductors and accessible parts SHALL be ≥2MΩ at 500V DC under dry conditions.	Must	Safety
E-015	Dielectric Strength	The device SHALL withstand a 1500V AC dielectric strength test between primary and secondary circuits for 1 minute without breakdown.	Must	Safety, Regulatory
E-016	Leakage Current	Touch current (leakage) SHALL not exceed 0.5mA (Class I device, EN 60335 criteria).	Must	Safety
E-017	Low-Voltage Supply (Internal)	Internal SMPS shall provide galvanically isolated 5V/12V/24V DC rails. Isolation SHALL meet reinforced insulation requirements per IEC 62368 or EN 60335.	Must	Safety
E-018	Over-Temperature Protection (Heater PCB)	A hardware independent thermal fuse (one-shot, ≤105°C trip) SHALL be physically mounted on or near each heating element as last-resort protection.	Must	Safety
E-019	Watchdog Timer	Firmware SHALL implement a hardware watchdog timer (≤1s timeout) to recover from software lockup. Safety-critical outputs (heaters) SHALL default to OFF on watchdog reset.	Must	Safety, Reliability
E-020	Battery Backup (RTC)	A coin cell battery SHALL maintain the real-time clock and incubation timer for a minimum of 5 years (device unused).	Must	Reliability
E-021	Motor Driver Protection	The egg-turning motor driver SHALL include overcurrent protection, stall detection, and soft-start to prevent mechanical and electrical damage.	Must	Reliability

A | APPEARANCE & AESTHETIC REQUIREMENTS

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
A-001	Enclosure Material	The main enclosure SHALL be constructed from ABS or similar food-safe, non-toxic plastic, resistant to cleaning agents (IPA, dilute bleach).	Must	Lifecycle, Safety
A-002	Colour and Finish	The exterior SHALL use neutral, easy-to-clean colour (white, light grey or beige). Surface finish shall be matte or satin to minimise fingerprint visibility.	Must	Aesthetics
A-003	Display Visibility	The LCD touchscreen SHALL be readable in ambient light conditions of 100–1000 lux. Minimum display contrast ratio: 300:1. Text size on main screen ≥ 14pt equivalent.	Must	Usability
A-004	Viewing Window	The incubation chamber SHALL include a transparent polycarbonate or tempered glass viewing window enabling egg observation without opening the lid.	Must	Usability, Aesthetics
A-005	LED Colour Temperature	Internal LED illumination SHALL be 5000K–6500K (daylight) for accurate egg candling observation.	Should	Function, Aesthetics
A-006	Touchscreen Bezel	The touchscreen shall have a smooth, sealed bezel (no gaps >0.5mm) to prevent ingress of debris and facilitate cleaning.	Must	Hygiene, Robustness
A-007	Cable Management	External cables (power cord, water tube) SHALL be routed via designated strain-relief fittings. No loose wiring shall be accessible to the user.	Must	Safety, Aesthetics

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
A-008	Controls Intuitiveness	The touchscreen UI SHALL require no more than 3 taps to reach any monitoring screen from the home screen. Icons SHALL be internationally recognisable.	Must	Usability
A-009	Door / Lid Design	The lid/door mechanism SHALL include a positive latch that requires deliberate action to open, preventing accidental opening during incubation.	Must	Usability, Safety
A-010	Label and Markings	All external labels SHALL be printed in UV-resistant inks or engraved. Labels SHALL remain legible for the full 10-year design life.	Must	Lifecycle, Regulatory
A-011	Product Dimensions	The product footprint SHALL accommodate up to 100 eggs in a standard tray configuration. Approximate target size: 600mm × 450mm × 350mm (W×D×H) ±15%.	Should	Function
A-012	Weight	The device fully loaded (eggs, water) SHALL not exceed 25kg to enable single-person handling.	Should	Ergonomics
A-013	Feet / Levelling	The device SHALL be equipped with 4 non-slip rubber feet. Levelling feet (adjustable) SHOULD be included to ensure horizontal placement.	Must	Function, Safety
A-014	Brand and Regulatory Markings	The product SHALL bear: CE mark, UKCA (if applicable), manufacturer logo, model number, mains rating, production date code, and WEEE symbol – all on the rear label.	Must	Regulatory

R | REGULATORY & RELIABILITY REQUIREMENTS

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
R-001	CE Marking	The product SHALL comply with all applicable EU Directives and bear CE marking prior to placing on the EU/EEA market (LVD, EMC, RoHS).	Must	Regulatory, Market
R-002	Low Voltage Directive (LVD)	Compliance with EU Directive 2014/35/EU is required. Essential safety requirements per EN 60335-1 and EN 60335-2 (household appliances) shall be met.	Must	Regulatory
R-003	EMC Directive	Compliance with EU Directive 2014/30/EU. EMC test reports shall be retained in the Technical File.	Must	Regulatory
R-004	RoHS Directive	All materials and components SHALL comply with EU RoHS 3 (Directive 2015/863/EU). Lead-free solder SHALL be used throughout.	Must	Regulatory, Environment
R-005	WEEE Directive	The product SHALL be registered under the WEEE Directive (2012/19/EU). WEEE symbol shall be marked on product. Take-back scheme must be in place for EU market.	Must	Regulatory, EOL
R-006	REACH Regulation	All substances of very high concern (SVHCs) in the product at >0.1% w/w shall be disclosed per REACH (EC No 1907/2006).	Must	Regulatory
R-007	Serbian Market (SRPS)	For the Serbian market, the product SHALL comply with applicable SRPS standards and be notified to the relevant authority. CE-equivalent declaration shall be prepared for Serbian authorities.	Must	Regulatory, Market

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
R-008	Technical Documentation	A complete Technical File (DoC, risk assessment, test reports, schematics, BoM, user manual) SHALL be compiled and retained for 10 years post last manufacture date.	Must	Regulatory
R-009	MTBF Target	The system MTBF SHALL be ≥20,000 hours under normal operating conditions (continuous operation, 37.8°C ambient chamber, 30°C external ambient).	Must	Reliability
R-010	Design Life	The product SHALL be designed for a minimum 10-year operational life with scheduled maintenance (cleaning, filter replacement) every 12 months.	Must	Reliability, Lifecycle
R-011	Heater Element Life	Heating elements SHALL be rated for ≥50,000 hours continuous duty or be designed as a replaceable field-serviceable module.	Must	Reliability
R-012	Fan Bearing Life	The circulation fan SHALL use sealed ball bearings rated for ≥30,000 hours MTBF.	Must	Reliability
R-013	Touchscreen Durability	The touchscreen SHALL be rated for ≥500,000 touch activations before failure.	Should	Reliability
R-014	Corrosion Resistance	All metallic fasteners and internal hardware SHALL be stainless steel (A2/304) or zinc-plated steel to resist corrosion from humid environment.	Must	Reliability, Lifecycle
R-015	Seal Degradation	Door gaskets and water system seals SHALL be silicone-based, resistant to temperature cycling (-20°C to +60°C), and rated for ≥10,000 open/close cycles.	Must	Reliability
R-016	Spare Parts Availability	The manufacturer SHALL commit to spare parts availability (fan, heater, humidity sensor, gaskets, control PCB) for a minimum of 7 years after end of production.	Must	Lifecycle, Service
R-017	Calibration	The temperature measurement chain SHALL be factory-calibrated and traceable to national standards. Calibration certificate SHALL be provided in the product documentation.	Must	Regulatory, Quality
R-018	Packaging	Product packaging SHALL comply with EU Packaging Directive (94/62/EC as amended). Packaging materials shall be recyclable and marked accordingly.	Should	Regulatory, Environment

S | SAFETY & STANDARDS REQUIREMENTS

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
S-001	Electrical Safety Standard	The device SHALL be designed and tested in accordance with EN 60335-1:2020 (General requirements for household electrical appliances) and applicable Part 2.	Must	Regulatory, Safety
S-002	Overheat Protection – Hardware	A hardware-independent thermal cutout (auto-reset thermostat ≤42°C + one-shot thermal fuse ≤105°C) SHALL be fitted in the heater circuit, independent of the microcontroller.	Must	Safety
S-003	Overheat Protection – Software	Firmware SHALL implement an independent high-temperature alarm and heater shut-off at 41.0°C. This	Must	Safety

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
		shall be a separate software task from the PID control loop.		
S-004	Runaway Heating Prevention	The system SHALL limit maximum heater duty cycle to prevent thermal runaway. If temperature rises >2°C above setpoint, heater shall immediately disable.	Must	Safety
S-005	Water Ingress Protection	The water and humidity systems shall be physically and electrically isolated from mains voltage. All water-carrying components shall be located away from mains wiring with a minimum 25mm creepage.	Must	Safety
S-006	Water Leak Detection	The system SHOULD incorporate a moisture/leak sensor at the base of the enclosure. A water leak SHALL trigger an alarm and disable the water dispensing pump.	Should	Safety
S-007	Burn Risk – Enclosure	No external surface accessible to the user shall exceed 60°C during normal operation (EN 60335 touch temperature limits for plastic surfaces).	Must	Safety
S-008	Child Safety	The device is intended for agricultural/hobbyist use by adults. Mains connection, water tank access, and lid opening SHALL require deliberate multi-step actions to mitigate child hazard. Choking hazard parts shall not be accessible.	Must	Safety
S-009	Ventilation	The device enclosure SHALL provide adequate ventilation for the electronics compartment to prevent overheating. Ventilation openings SHALL be designed to prevent ingress of eggs, debris, or fingers (IP2X for finger probe).	Must	Safety
S-010	Moving Parts Guard	The egg-turning mechanism and fan shall be guarded so that no user-accessible opening allows finger contact with moving parts (EN 60335 finger probe test).	Must	Safety
S-011	Power Cord	The mains power cord SHALL be a certified HAR-marked flexible cable ≥H05VV-F 3G1.0mm², minimum 1.5m long, fitted with a CEE 7/7 schuko plug for Balkan/EU markets.	Must	Safety, Regulatory
S-012	Abnormal Operation	The device SHALL be tested for abnormal conditions per EN 60335-1 Clause 19: blocked ventilation, locked fan, heater short, sensor open/short. No fire, shock, or unsafe emission shall result.	Must	Safety
S-013	Malicious/Incorrect Use – Overfill	If water tank is overfilled, the system SHALL detect overflow via water level sensor and cease pumping before overflow reaches electronics.	Must	Safety, Misuse
S-014	Malicious/Incorrect Use – Wrong Voltage	The device SHALL incorporate a mains voltage selector or be rated for 230V ±10% only. Connecting to 110V SHALL not cause damage; connecting to higher voltage SHALL be mitigated by MOV/fuse protection.	Should	Safety, Misuse
S-015	Malicious Firmware	Firmware update procedure SHALL validate image authenticity (CRC + signing) before flashing. Unsigned or corrupted firmware SHALL be rejected and current firmware retained.	Must	Security, Misuse
S-016	Single Fault Safety	Under any single electrical fault (short circuit, open circuit, component failure), the device SHALL not present a risk of fire, electric shock, or dangerous temperature.	Must	Safety

ID	Sub-System / Feature	Requirement Statement	Priority	Use Case / Context
S-017	Humidity Chemical Safety	The water used for humidity generation SHALL be plain water only. Product documentation SHALL explicitly prohibit use of chemicals or additives in the water tank.	Must	Safety
S-018	Biological Decontamination	The enclosure interior and accessible surfaces SHALL be designed to withstand decontamination with 70% IPA or 1% sodium hypochlorite solution without material degradation.	Must	Hygiene, Lifecycle
S-019	End-of-Life Disposal	The product SHALL be labelled with WEEE symbol. User documentation SHALL provide instructions for safe disposal and identify components requiring special handling (battery, PCB).	Must	Regulatory, EOL
S-020	Electromagnetic Compatibility – Safety Devices	Alarm circuits and safety cutouts SHALL be immune to EMI and shall function correctly even during worst-case EMC test conditions.	Must	Safety, Regulatory
S-021	Shipping / Drop Safety	The unit (packaged) SHALL survive a 0.5m drop onto a concrete surface without permanent functional damage or safety compromise.	Should	Lifecycle

APPENDIX A | APPLICABLE STANDARDS & DIRECTIVES

The following standards and directives are applicable to this product. Mandatory standards must be demonstrated through testing and/or technical documentation for market access. Recommended standards represent best practice.

Standard / Directive	Title / Scope	Category	Status
EN 60335-1:2020 + A11:2022	Safety of household and similar electrical appliances – Part 1: General requirements	Safety	Mandatory (LVD)
EN 60335-2-101	Safety – Part 2-101: Particular requirements for vaporisers (humidity)	Safety	Applicable
IEC 62368-1:2018 (EN 62368-1)	Audio/video, IT and comms – Safety requirements (supersedes EN 60065 / EN 60950)	Safety	Applicable if classified under AV/IT
EN 55032:2015+A1:2020	Electromagnetic compatibility – Emission requirements for multimedia equipment	EMC	Mandatory (EMC Directive)
EN 55035:2017+A11:2020	Electromagnetic compatibility – Immunity requirements for multimedia equipment	EMC	Mandatory (EMC Directive)
IEC 61000-4-2:2009	ESD immunity test	EMC/Immunity	Mandatory
IEC 61000-4-4:2012	EFT/Burst immunity test	EMC/Immunity	Mandatory
IEC 61000-4-5:2017	Surge immunity test	EMC/Immunity	Mandatory
IEC 61000-4-11:2020	Voltage dips and interruptions immunity	EMC/Immunity	Mandatory
EN IEC 61000-3-2:2019+A1:2021	Harmonic current emissions (≤16A)	EMC	Mandatory
EN 61000-3-3:2013+A1:2019	Voltage fluctuations and flicker (≤16A)	EMC	Mandatory

Standard / Directive	Title / Scope	Category	Status
EU Directive 2014/35/EU	Low Voltage Directive (LVD)	Regulatory	Mandatory for CE
EU Directive 2014/30/EU	EMC Directive	Regulatory	Mandatory for CE
EU Directive 2011/65/EU + 2015/863/EU	RoHS 3 – Restriction of Hazardous Substances	Regulatory/Environmental	Mandatory
EU Directive 2012/19/EU (WEEE)	Waste Electrical and Electronic Equipment Directive	Regulatory/EOL	Mandatory
EC Regulation 1907/2006 (REACH)	Registration, Evaluation, Authorisation of Chemicals	Regulatory/Environmental	Mandatory
EU Directive 94/62/EC (Packaging)	Packaging and packaging waste	Regulatory/Environmental	Applicable
SRPS EN 60335-1	Serbian national adoption of EN 60335-1 (safety)	Regulatory – Serbia	Mandatory for Serbian market
SRPS EN 55032 / SRPS EN 55035	Serbian national adoption of EMC standards	Regulatory – Serbia	Mandatory for Serbian market
EN IEC 63000:2018	Technical documentation for RoHS compliance assessment	Documentation	Recommended
IEC 60068-2-1/2/14	Environmental testing – Cold, Dry heat, Temperature change	Environmental/Reliability	Recommended
IEC 60068-2-30	Environmental testing – Damp heat cyclic	Environmental/Reliability	Recommended
IEC 60529	Degrees of protection (IP code) – Ingress protection	Safety/Environmental	Mandatory (IP rating claim)
EN ISO 9241-11:2018	Ergonomics of human-system interaction – Usability	Usability	Recommended
IEC 62133	Battery safety (if rechargeable battery fitted for backup)	Safety	Applicable if Li-ion used
EN 301 489-1 (if wireless)	EMC for radio equipment – Common technical requirements	EMC/Radio	If wireless functionality added
EN 300 328 (if 2.4GHz WiFi/BT)	Radio equipment – 2.4GHz wideband transmission	Radio/EMC	If wireless functionality added

APPENDIX B | EDGE CASES & MISUSE SCENARIOS ADDRESSED

The following edge cases, wrong uses, and malicious uses have been explicitly addressed within the requirements above:

Scenario	Risk	Requirement(s) Addressing It
Accidental factory reset	Loss of all incubation settings mid-cycle, potential egg loss	F-018
Power failure mid-incubation	Loss of timer, settings, and temperature control	F-016, E-020
Sensor failure (open/short)	Undetected temperature runaway or incorrect readings	F-002, F-004, F-020, S-012

Scenario	Risk	Requirement(s) Addressing It
Fan blockage / stall	Uneven temperature, overheating, egg loss	F-007, E-021
Water tank dry	Humidity collapse, embryo desiccation	F-005
Water tank overfill / leak	Electrical shock hazard, electronics damage	S-005, S-006, S-013
Child access to device	Burn risk, electric shock, ingestion of parts	S-008, S-009, S-010
User pours chemicals into water tank	Chemical vapour inside chamber, health hazard	S-017
Device used at wrong mains voltage	Component damage, fire, shock hazard	S-014, E-009
Malicious or corrupted firmware upload	Loss of safety functions, data exfiltration	F-025, S-015
Door opened accidentally during incubation	Temperature/humidity loss, embryo mortality	A-009, F-012
Unauthorised parameter changes	Temperature/humidity set outside safe range	F-022
Software lockup / crash	Heater left on or fan disabled indefinitely	E-019
Device dropped during installation	Structural and safety failure	S-021
Condensation inside electronics compartment	PCB corrosion, short circuit, shock hazard	E-012, E-013
End of life – improper disposal	Toxic materials entering waste stream	R-005, R-006, S-019
Continued use after thermal fuse trip	User bypasses protection; fire/shock risk	E-018, user documentation, fuse access design

Document End – Requirements Engineering Team